

Constant acceleration kinematics (1)

$$\vec{v}(t) = \vec{v}_0 + \vec{a}t \quad (2)$$

$$\vec{r} = \vec{r}_0 + \vec{v}_0 t + \frac{1}{2} \vec{a}t^2 \quad (3)$$

(4)

uniform circular motion (5)

$$\vec{v} = \omega r \quad (6)$$

$$a_c = \omega^2 R = \frac{v^2}{R} \quad (7)$$

$$\omega = \frac{2\pi}{T} \quad (8)$$

Momentum (9)

$$\vec{F}_{net} = \frac{d\vec{p}}{dt} \quad (10)$$

$$\vec{p} = m\vec{v} \quad (11)$$

$$\vec{p}_{tot} = \sum \vec{p}_i \quad (12)$$

$$\vec{p}_{tot} = M_{tot} \vec{v}_{cm} \quad (13)$$

$$\vec{p}_{cm} = M_{tot} \vec{v}_{cm} \quad (14)$$

Center of Mass : (15)

$$\vec{R}_{cm} = \frac{\sum_{i=1}^N m_i \vec{r}_i}{M_{tot}} \quad (16)$$

$$\vec{R}_{cm} = \frac{\int \vec{r} dr}{m_{tot}} \quad (17)$$

Forces: (18)

$$F_{grav} = ma \quad (19)$$

$$F_{spr} = k\Delta x \quad (20)$$

$$\mu = \frac{F_f}{F_n} \quad (21)$$

Energy: (22)

$$E_{tot} = \frac{1}{2}mv^2 + mgY \quad (23)$$

$$U_{grav} = mgY \quad (24)$$

$$U_{spring} = \frac{1}{2}kx^2 \quad (25)$$

(26)

For an object rotating about a fixed axis,

$$K = \frac{1}{2}I\omega^2$$

for an object rotating about it's center of mass, with the center of mass rotating at speed V_{cm} ,

$$K = \frac{1}{2}I'\omega^2 + \frac{1}{2}MV_{cm}^2$$

Center axis theorem

$$I = I' + Md^2$$

Angular momentum:

$$\vec{L} = I' \vec{\omega} + \vec{R}_{cm} \times \vec{P}_{cm}$$

For more on forces, see *Newtons Laws*